NeuroInsight report

Turning visual stimuli into neuroscience-informed insights | 1200x800 px | analysis 2026-10-11T16:15:58+00:00

Executive summary

Computed directly from this image: a luminance contrast ratio of **17.38:1** (AAA), an edge density of **0.9%**, and a mean luminance of **0.61**. Predicted attention peaks in region **R2C1** holding **15.6%** of the modelled attention. Face detection returned **0** face(s); OCR recovered **3** text block(s).

Everything below is tagged by how it was established. Measured values are facts about the image; model outputs are predictions; literature entries are general mechanisms that are not specific to this image; hypotheses are our reading. No claim is made about any individual's response.

Attention map



Predicted saliency overlay (turbo colour scale, warm = higher predicted attention). This is a computational estimate from OpenCV Static Saliency (Spectral Residual + Fine-Grained). **No person was observed viewing this image.** It is not eye-tracking and must not be presented as measured gaze.

Findings

PREDICTED BY A PRETRAINED MODEL

Predicted attention concentrates in a moderate region

The saliency model places its peak in region R2C1 (row 2, column 1), which carries 15.6% of total predicted attention and sits at a mean luminance of 213/255. Attention is predicted to spread over most of the frame, so no single element can be assumed to dominate.

Confidence: Moderate — model-dependent, validated on saliency datasets | refs: koehler2016, itzel2017

ESTABLISHED MECHANISM IN THE LITERATURE

This is predicted attention, not measured gaze

The map above is a computational estimate from OpenCV Static Saliency (Spectral Residual + Fine-Grained). No person was observed viewing this image. Predicted saliency is a useful prior about where attention tends to fall, and it correlates with fixation data, but it is not a substitute for eye-tracking and should never be reported as measured gaze.

Confidence: Moderate-High — replicated, but not specific to this image | refs: koehler2016, itzel2017

DIRECTLY CALCULATED FROM YOUR IMAGE

Luminance contrast ratio 17.38: AAA (>=7:1) — passes WCAG for normal text

Brightest and darkest regions differ by a WCAG relative-luminance ratio of 17.38:1 (dynamic range 99.0% of the 0–1 luminance scale). This comfortably exceeds the 7:1 threshold at which WCAG defines maximum contrast compliance, so the image's most and least luminous elements should be readily distinguishable. Note this measures the extremes of the image, not whether specific text is legible in context.

Confidence: High — recomputable from the image

DIRECTLY CALCULATED FROM YOUR IMAGE

Visual complexity is low

Edge density is 0.9% of pixels and the luminance distribution has an entropy of 1.64 bits (8.0 = perfectly uniform). The image presents a moderate number of elements. Attention is less likely to be swamped, though some regions will still be missed in brief exposure.

Confidence: High — recomputable from the image | refs: bellantoni2008

PREDICTED BY A PRETRAINED MODEL

No faces detected

The cascade detector found no faces. This is a negative detection, not proof of absence: the Viola-Jones cascades are biased toward frontal, well-lit faces and miss profile, small and angled faces.

Confidence: Moderate — model-dependent, validated on saliency datasets

PREDICTED BY A PRETRAINED MODEL

3 text blocks detected, largest is "BOOST"

OCR recovered 4 words at a mean confidence of 0.83. The largest block covers 2.319% of the frame. Copy occupying a larger share of the field is more likely to be read, and typographic weight is one of the stimulus properties that drives initial orienting of attention.

Confidence: Moderate — model-dependent, validated on saliency datasets | refs: zhou2015

DIRECTLY CALCULATED FROM YOUR IMAGE

Dominant hue at 8° covering 87.8% of chromatic pixels

Mean saturation is 51/255 and 3270.0% of pixels carry meaningful chroma. A dominant hue concentrates visual identity and makes an image easier to group as a single object; strong colour contrast between an element and its background raises the probability that the element is separated early during visual search.

Confidence: High — recomputable from the image | refs: treisman1980

NOT DETERMINABLE FROM A STATIC IMAGE

Cannot be determined: the viewer's emotional response

A static image cannot support a claim about the viewer's emotional response. Doing so would require participant-level data such as a validated affect rating scale. We report this gap explicitly instead of estimating it, because a fabricated figure would be worse than no figure.

Confidence: None — stated explicitly as not answerable

NOT DETERMINABLE FROM A STATIC IMAGE

Cannot be determined: purchase intent or conversion probability

A static image cannot support a claim about purchase intent or conversion probability. Doing so would require participant study with a predefined outcome measure and a trained model. We report this gap explicitly instead of estimating it, because a fabricated figure would be worse than no figure.

Confidence: None — stated explicitly as not answerable

NOT DETERMINABLE FROM A STATIC IMAGE

Cannot be determined: neural activity in the brain

A static image cannot support a claim about neural activity in the brain. Doing so would require fMRI, MEG or EEG recording of a person viewing the stimulus. We report this gap explicitly instead of estimating it, because a fabricated figure would be worse than no figure.

Confidence: None — stated explicitly as not answerable

Recommendations (hypotheses to test)

These are testable predictions, not proven improvements. Each requires a controlled comparison with participant data before it could be called an improvement.

[LOW] With no face detected and text carrying the attention, the visual hierarchy may be carried by typography alone, which limits the social-affective route to attention.

How to test it: Compare a typographic-only variant against one adding a face. Measure fixation latency and dwell time on the product.

Limitations

- Classical (non-deep) method; no learned representation of faces or text.
- Optimised for natural photographs, not designed advertisements.
- Strongly driven by contrast; can miss semantically salient but low-contrast regions.
- Trained prior to modern ad-design practice.
- Biased toward frontal, well-lit faces; under-detects profile and angled faces.
- Fixed-scale training; accuracy drops on small or low-resolution faces.
- Produces bounding boxes only — no identity, no expression, no demographics.
- Higher false-positive rate on textured backgrounds.
- Confidence reflects OCR legibility, not persuasive intent.
- Small or low-contrast text is frequently missed.
- Stylised display type is harder than body copy.

This tool estimates properties of *images*. It cannot measure a person. It is a research prototype: it is not clinically validated, not a commercial neuromarketing system, and not a substitute for fMRI, eye-tracking or field experiments. Predicted attention should be

treated as a hypothesis to be tested, never as a finding about an audience.

References

Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. Cognitive Psychology, 11(2), 333–356. doi:10.1016/0010-0285(80)90005-5
Supports: Binding theory: attention combines features (colour, orientation, motion, shape) into coherent objects; conjunction search requires focused attention.

Itzel, H. M., Keller, B. A., & Zitelli, F. (2017). Mathematical modeling of the role of prefrontal cortex for the emergence of visual saliency. Scientific Reports, 7, 41100. doi:10.1038/s41598-017-04161-0
Supports: Visual saliency has a measurable neural basis in frontoparietal circuitry; saliency is computed, not purely a property of stimulus energy.

Koehler, W., Kower, M. J., Salscheuer, D., Stalder, S. S., & Friedrich, J. (2016). Saliency of natural scenes revisited. Journal of Vision, 16(2), 34. doi:10.1167/jov.16.2.34
Supports: Empirical measurement of where observers look in natural scenes; saliency is driven by local contrast, task, and scene context, not a single feature.

Zhou, Y., Boyd, A. P., DeForest, J. R., & Meschery, J. (2015). Self-referential aesthetic and cognitive appeals to visual attention and image categorization. Psychonomic Bulletin & Review, 22(4), 760–766. doi:10.3758/s13423-015-0407-5
Supports: Typographic weight and spatial arrangement change where attention is first directed in an image, even holding colour content constant.

Swellens, J., Poels, K., & Wagemans, J. (2022). The face-size effect in natural images: an efficient coding approach. Psychological Research, 86, 1437–1451. doi:10.1007/s00426-021-02236-0
Supports: Larger face-like regions in an image capture fixation more reliably; face presence biases attention disproportionately to their area.

Bellantoni, M., & Oliva, A. (2008). Pursuing the informational value of picture in advertisements: the role of attentional bias and cognitive load. Journal of Advertising, 37(2), 2–15. doi:10.1080/00913367.2008.1242148
Supports: High cognitive load reduces the processing of brand and product information in advertisements.

Mandler, J. M. (1991). Cognition. In J. R. Anderson (Ed.), Cognitive Psychology and Its Implications (3rd ed.).
Supports: Perceptual organisation: people represent scenes as hierarchies of parts within wholes, which is why focal regions matter.

Hou, X., & Zhang, L. (2007). Saliency detection: A spectral residual approach. Proceedings of CVPR 2007, 1–8. doi:10.1109/CVPR.2007.383289
Supports: Describes the spectral-residual saliency method implemented in this project.

Generated by NeuroInsight by Aanishka v0.1.0 on 2026-10-11 21:47.